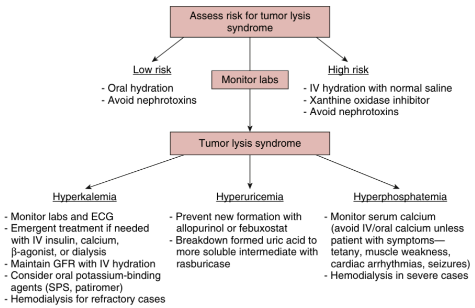

Fig. 57.1 Approach to the management of tumor lysis syndrome. *ECG,* Electrocardiogram; *GFR,* glomerular filtration rate; *IV,* intravenous; *SPS,* sodium polystyrene sulfonate. (From Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med.* 2017;376(18):1770–1781.)

Fig_57_1

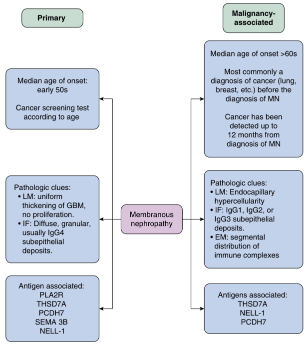

Fig. 57.2 Approach to distinguish primary membranous nephropathy from cancer-associated membranous nephropathy. *EM,* electron microscopy; *GBM,* glomerular basement membrane; *IF,* immunofluorescence; *Ig,* immunoglobulin; *LM,* light microscopy; *MN,* membranous nephropathy; *NELL-1,* neural epidermal growth factor–like 1 protein; *PCDH7,* protocadherin 7; *PLA2R,* phospholipase A2 receptor; *SEMA 3B,* semaphorin-3B; *THSD7A,* thrombospondin type I domain-containing 7A.

Fig_57_2

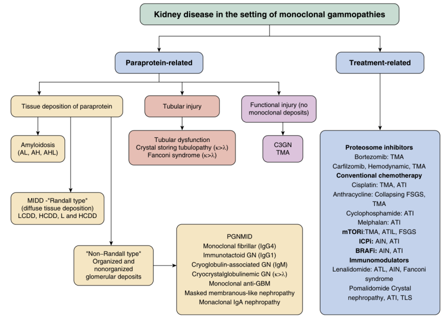

Fig. 57.3 An approach to the differential diagnosis of kidney disease associated with paraproteinemias. *AH,* Amyloid heavy chain; *AHL,* amyloid light and heavy chain; *AIN,* acute interstitial nephritis; *AL,* amyloid light chain; *ATI,* acute tubular injury; *BRAFi,* v-raf murine sarcoma viral oncogene homolog B1 inhibitor; *C3,* complement 3; *FSGS,* focal segmental glomerulosclerosis; *HCDD,* heavy chain deposition disease; *ICPi,* immune checkpoint inhibitor; *Ig,* immunoglobulin; *IgG 1&4,* immunoglobulin G types 1 and 4; *IgM,* immunoglobulin M; *LCDD,* light chain deposition disease; *LHCDD,* light-heavy chain deposition disease; *MIDD,* monoclonal immunoglobulin deposition disease; *mTORi,* mammalian target of rapamycin inhibitor; *PGMIDD,* proliferative N with monoclonal IgG deposits; *TLS,* tumor lysis syndrome; *TMA,* thrombotic microangiopathy; *TNF,* tumor necrosis factor.

Fig_57_3

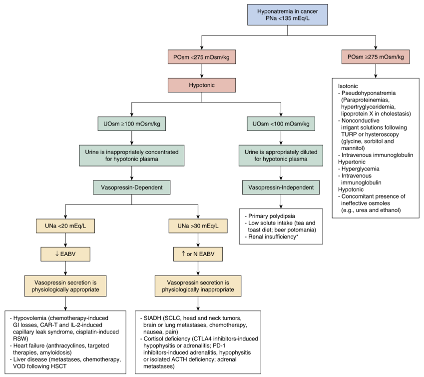

Fig. 57.4 Diagnostic approach to hyponatremia in the patient with cancer.

Fig_57_4

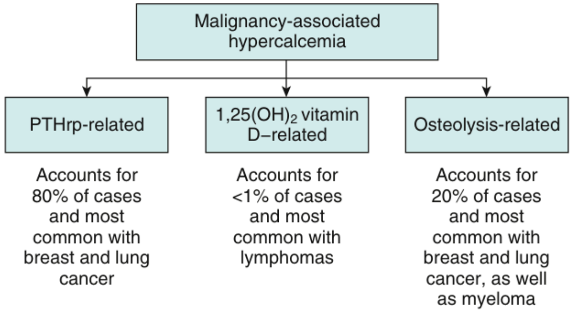

Fig. 57.5 Causes of malignancy-associated hypercalcemia. *PTHrP,* Parathyroid hormone related peptide.

Fig_57_5

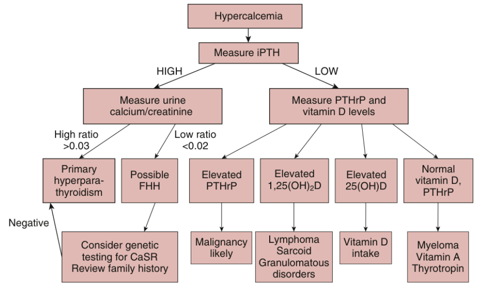

Fig. 57.6 Diagnostic algorithm for the patient presenting with hypercalcemia. Measurement of PTH is the first step in the diagnostic approach. Patients with malignancy-associated hypercalcemia typically have low levels of PTH due to feedback inhibition of the high blood calcium levels on the parathyroid gland. In patients with low levels of PTH, measurement of PTHrP and 1,25-dihydroxyvitamin D (1,25(OH)$_2$D) levels will allow classification of patients into several diagnostic groupings. In patients with elevated PTH levels, a 24-hour urine collection for calcium and creatinine levels should be obtained. *CaSR,* Calcium-sensing receptor; *FHH,* familial hypocalciuric hypercalcemia; *iPTH,* intact parathyroid hormone; *PTH,* parathyroid hormone; *PTHrP,* parathyroid-related peptide; *25(OH)D,* 25-hydroxy vitamin D. (From Reagan P, Pani A, Rosner MH. Approach to diagnosis and treatment of hypercalcemia in a patient with malignancy. *Am J Kidney Dis.* 2013;63(1):141–147.)

Fig_57_6

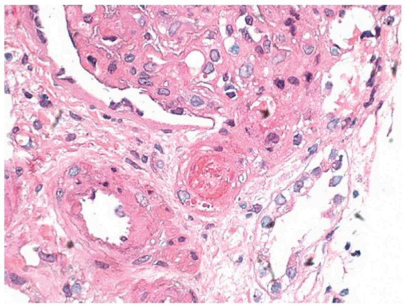

Fig. 57.7 Kidney biopsy from a patient with thrombotic microangiopathy in the setting of hematopoietic stem cell transplantation exhibits a fresh fibrin thrombus in a preglomerular arteriole. (Hematoxylin-eosin, ×600.)

Fig_57_7

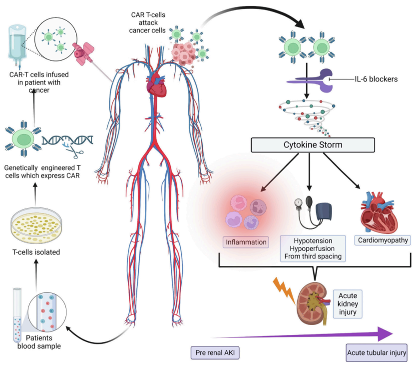

Fig 57.8 Mechanism of how chimeric antigen receptor T cell therapy works and may lead to cytokine release storm and acute kidney injury.

Fig_57_8

Table 57.1 Risk Factors for Acute Kidney Injury in Patients With Cancer

Modifiable Risks	Nonmodifiable Risks
Nephrotoxic medications (nonchemotherapeutic), such as aminoglycosides, amphotericin B, calcineurin inhibitors	Age >65 years
Nephrotoxic medications (chemotherapeutic), such as cisplatin, checkpoint inhibitors, ifosfamide, interferon, tyrosine kinase inhibitors, BRAF (serine–threonine protein kinase) inhibitors	Underlying chronic kidney disease, especially diabetic kidney disease
Hypovolemia • True volume depletion—nausea, vomiting, diarrhea • Effective volume depletion—cirrhosis, heart failure, nephrotic syndrome, hypoalbuminemia	Specific cancer types—multiple myeloma, hepatocellular cancer, renal cell carcinoma, pelvic malignancies
Intravenous radiographic contrast	Hematopoietic stem cell transplantation
High tumor bulk and risk for tumor lysis syndrome	Female gender

Table_57_1

Table 57.2 Intrarenal and Postrenal Causes of Acute Kidney Injury in the Patient With Cancer

Cause	Representative Example
Vascular—thrombotic microangiopathy	Gemcitabine, post stem cell transplantation
Tubular injury—acute tubular necrosis	Ischemia due to sepsis, nephrotoxicity due to chemotherapeutic agents such as cisplatin
Tubular injury—intratubular precipitation of crystals	Methotrexate
Tubular injury—cast nephropathy	Multiple myeloma
Tubular injury—lysozymuria	Acute promyelocytic, acute monocytic leukemia, chronic myelomonocytic leukemia
Interstitial injury—interstitial nephritis	Immune checkpoint inhibitors
Interstitial injury—tumor infiltration of the kidneys	Lymphoma
Glomerular injury	Various types of amyloidosis Various paraneoplastic or drug-induced glomerulonephritis Paraneoplastic membranous nephropathy
Obstruction	Retroperitoneal lymphadenopathy (lymphoma) Tumor bulk obstructing urine flow (pelvic malignancies)

Table_57_2

Table 57.3 Cairo-Bishop Criteria for Tumor Lysis Syndrome

Type of Criteria	Features
Laboratory criteria for tumor lysis syndrome	Requires 2 or more of the following criteria within 3 days to or 7 days after initiation of chemotherapy: • Uric acid level ≥8 mg/dL or 25% increase from baseline • Potassium level ≥6.0 mEq/L or 25% increase from baseline • Phosphorus level ≥6.5 mg/dL for children • Phosphorus level ≥4.5 mg/dL for adults or 25% increase from baseline • Calcium level ≤7 mg/dL or 25% decrease from baseline
Clinical criteria for tumor lysis syndrome	Laboratory tumor lysis syndrome plus one or more of the following clinical criteria: • Creatinine >1.5 times upper limit of age-adjusted reference range • Cardiac dysrhythmia or sudden death • Seizure

From Cairo MS, Bishop M. Tumor lysis syndrome: new therapeutic strategies and classification. *Br J Hematol*. 2004;127(1):3–11.

Table_57_3

Table 57.4 Standardized Incidence Rates of Cancer in Hemodialysis Patients[a]

Type of Cancer	Standardized Incidence Rate
All cancer types	1.42
Kidney parenchyma/pelvis	4.03
Bladder	1.57
Breast (female)	1.42
Non-Hodgkin lymphoma	1.37
Pulmonary, lung	1.28
Colonic, rectal	1.27
Pancreatic	1.08
Prostatic	1.06

[a]See references.[55-61]

Table_57_4

Table 57.5 Dose Modifications for Commonly Used Chemotherapeutic Agents in Chronic Kidney Disease Patients[69,75,76]

	Creatinine Clearance (mL/min)			
Agent	**90-60**	**60-30**	**30-15**	**Hemodialysis**
Bleomycin	100%	75%	75%	50%[a]
Capecitabine	100%	75%	Avoid	Avoid
Carboplatin	Dosing based on AUC Calvert formula—total carboplatin dose (mg) = target AUC × (eGFR + 25); AUC varies between 5 (treated pts) and 7 (untreated pts; mg/mL × min)			In dialysis patients, consider GFR = 0 and target dose = 125–175 mg; *E*, 84% ± 3%[b]
Carmustine	100%	80% if CrCl ≥45; 75% if CrCl <45	Consider alternative	Consider alternative
Cisplatin	100%	50%	50%	25%-50%[b]
Crizotinib	100%	100%	50%	NA
Cyclophosphamide	100%	75%-100%	75%	75%a; *E*, 40%-90%
Cytarabine (high dose)	100%	60% if CrCl ≥45; 50% if CrCl <45	30% or consider alternative	30% consider alternative[a]
Dacarbazine	100%	80%-70%	NA	100 mg daily for 5 days per cycle
Eribulin	100%	≥40 mL/min: 100%; <40 mL/min—NA	NA	NA
Etoposide	100%	75%	75%	50%[c]
Fludarabine	100%	80%, United States; 50%, United Kingdom	Avoid	Avoid
Hydroxyurea	100%	50%	50%	20%[b]
Ifosfamide	100%	75%-100%	75%-100%	75%[b]; *E*, 87%
Irinotecan	100%	NA	NA	30%-40%
Lenalidomide	25 mg daily	10 mg daily	15 mg every 2 days	15 mg every 2 days; *E*, 31%[b]
Lomustine	100%	75%	75%	25%-50%
Melphalan	100%	75%	75%	50%
Methotrexate	100%	80%	50%	Avoid; HD Cl, 92 ± 10 mL/min[b]
Mitomycin	100%	100%	100%	75%[a]
Oxaliplatin	100%	100%	Avoid if CrCl <30 (Canada) or <20 (United States)	Avoid
Pentazocine	100%	66%	66%	50%
Pentostatin	100%	50%-75%	50%	50% (1-2 h before dialysis)
Pemetrexed	100%	100%	Avoid	Avoid
Regorafenib	100%	100%	NA	NA
Sorafenib	100%	100%	100%	25% of the dose; increase to 100% according to clinical safety and efficacy[c]
Sunitinib	100%	100%	100%	25%; then increase to 100%, according to clinical safety and efficacy; *E*, 0[c]
Topotecan	100%	50%	50%	25%[b]
Vandetanib	100%	25%	NA	NA

[a]In the absence of data on its removal during dialysis, the drug will be administered after the session.
[b]The drug is dialyzable. Therefore it will be administered after the session, on days with HD.
[c]The drug may be administered indifferently, before or after the HD session.

Table_57_5

Table 57.6 Glomerular Diseases Associated With Cancer and Chemotherapeutic Agents

Glomerular Lesion	Malignancy	Drug
Membranoproliferative glomerulonephritis	Lung, renal cell, breast, esophageal, gastric, Wilms, melanoma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, monoclonal gammopathy, myeloma	Gemcitabine, sirolimus, immune checkpoint inhibitor
Minimal change disease	Lung, colon, pancreas, bladder, renal cell, ovarian, mesothelioma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Interferon-α, β, γ; pamidronate, doxorubicin, daunorubicin, sirolimus, immune checkpoint inhibitors, tyrosine kinase inhibitors
Focal segmental glomerulosclerosis (FSGS)	Lung, renal cell, breast, esophageal, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, T cell leukemia	Sirolimus, temsirolimus, everolimus, doxorubicin, daunorubicin, bisphosphonates, tyrosine kinase inhibitors
Collapsing FSGS	Unlikely, but causes of FSGS listed above could be considered	Interferon-α, β, γ; pamidronate, gefitinib, sirolimus, doxorubicin, daunorubicin, clofarabine, bisphosphonates
Membranous nephropathy	Lung, colon, gastric, pancreas, prostate, breast, head and neck, Wilms, teratoma, ovarian, cervical, endometrial, melanoma, skin, pheochromocytoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Immune checkpoint inhibitors
Lupus-like glomerulonephritis	None	Immune checkpoint inhibitors
IgA nephropathy	Lung, pancreas, renal cell, head and neck, Hodgkin and non-Hodgkin lymphoma	Sirolimus, immune checkpoint inhibitors

Table_57_6

Table 57.7 Causes of Hypokalemia in the Patient With Cancer

Type	Cause
Tumor-related	Lysozymuria with acute leukemia Mineralocorticoid excess syndromes • Primary hyperaldosteronism (adrenal carcinoma) • Renin-producing tumors (extremely rare) • Ectopic adrenocorticotropin syndrome Intracellular shifts (pseudohypokalemia) in the setting of high white blood cell counts
Chemotherapy-related	Chemotherapeutic agents (cisplatin, ifosfamide) Antimicrobials and antifungal agents (aminoglycosides, amphotericin B) Excessive gastrointestinal losses (vomiting, diarrhea)
Other related	• Hypercalcemia • Hypomagnesemia • Postobstructive diuresis • Use of growth factors

Table_57_7

Table 57.8 Dosing Guidelines for Use of Bisphosphonates in Treatment of Malignancy-Associated Hypercalcemia

Bisphosphonate	Dosing Guidelines	Interval Between Doses
Pamidronate	CrCl >60 mL/min: 90 mg IV administered over 2-3 hours CrCl, 30-60 mL/min: 60-90 mg IV administered over 2-4 hours CrCl < 30 mL/min: 60-90 mg IV administered over 4-6 hours	Every 3-4 weeks
Zoledronate	CrCl >60 mL/min: 4 mg IV over 15 minutes CrCl, 50-60 mL/min: 3.5 mg IV over 15 minutes CrCl, 40-49 mL/min: 3.3 mg IV over 15 minutes CrCl 30-49 mL/min: 3 mg IV over 15 minutes CrCl <30 mL/min: not recommended	Every 3-4 weeks

Table_57_8

Table 57.9 Risk Factors and Causes of Acute and Chronic Kidney Lesions With Hematopoietic Stem Cell Transplantation

Acute Kidney Injury	Chronic Kidney Disease
Prerenal—nausea, vomiting, and diarrhea associated with acute gastrointestinal GVHD, drug-induced nausea and vomiting	Thrombotic microangiopathy—calcineurin inhibitors, chronic GVHD
Prerenal/acute tubular injury—sepsis, sinusoidal obstruction syndrome, acute GVHD, marrow infusion syndrome	Arteriolonephrosclerosis—hypertension, radiation, TBI
Thrombotic microangiopathy—acute GVHD, calcineurin inhibitors, TBI	Viral nephropathy—BK virus, cytomegalovirus
Acute tubular injury, crystalline nephropathy—amphotericin, vancomycin, aminoglycosides, acyclovir, other nephrotoxins	Membranous nephropathy, minimal change disease—chronic GVHD, drug-induced glomerular injury

GVHD, Graft-versus-host disease; *TBI*, total body irradiation.

Table_57_9

Table 57.10 Anticancer Drug Nephrotoxicity

Drug Class	Drug Name	Mechanism of Injury	Kidney Effect
Conventional Chemotherapy			
Platinum based	Cisplatin, Carboplatin, Oxaliplatin.	Platinum-DNA adducts mediate arrest of cell cycle, initiate apoptosis. ATP depletion	ATI, renal magnesium wasting, proximal tubulopathy, NDI
Antimetabolite	Methotrexate	Intratubular crystal formation. Afferent arteriolar constriction	Crystal nephropathy, ATI
	Gemcitabine	Endothelial injury	TMA, HTN
	Pemetrexed	Unknown	ATI, Chronic interstitial fibrosis, proximal tubulopathy, NDI
Alkylating agent	Cyclophosphamide	Toxic metabolite, acrolein	hemorrhagic cystitis
	Ifosfamide	Toxic metabolite, chloroacetaldehyde	ATI, proximal tubulopathy, NDI
	Melphalan	Increase ADH release	SIADH
	Nitrosoureas	Alkylation of tubular cell proteins	Chronic interstitial nephritis
Antitumor antibiotics	Mitomycin C	Endothelial injury	TMA, HTN
Targeted Therapy			
VEGF inhibitors	Bevacizumab	Endothelial injury	TMA, HTN, ATI
Tyrosine kinase inhibitors	Sorafenib, sunitinib, axitinib, pazopanib, lenvatinib.	Endothelial injury, podocyte injury.	ATIN, ATI, TMA, FSGS
BCR-ABL tyrosine kinase inhibitor	Imatinib, dasatinib.	Tubular injury, endothelial injury	ATI, TMA
BRAF inhibitors	Vemurafenib, dabrafenib	Tubular injury, ERK activation.	ATIN, ATI
BCL-2 inhibitors	Venetoclax	Tubular injury	AKI, TLS
ALK inhibitors	Crizotinib,lorlatinib, alectinib	Inhibition of creatinine secretion, renal arteriolar myocyte vacuolization	Pseudo-AKI, ATIN, ATI, kidney cyst, podocytopathies
CDK4/6 inhibitors	Palbociclib, ribociclib	Inhibit MATE1 and MATE2 transporters	Pseudo-AKI, ATI
PARP inhibitors	Olaparib, talazoparib	Inhibition of creatinine secretion	Pseudo-AKI
MET tyrosine kinase	Capmatinib, tepotinib	Inhibition of creatinine	Pseudo-AKI
EFGR monoclonal antibodies	Cetuximab, panitumumab	Inhibition of EGFR signaling at the DCT	Renal magnesium wasting
mTOR inhibitors	Everolimus	Decrease cubilin and megalin, VEGF inhibition	ATI, podocytopathies
Protease inhibitors	Bortezomib, carfilzomib	Endothelial injury, autoantibody formation	TMA, HTN
Bruton tyrosine kinase inhibitors	Ibrutinib	Endothelial injury	ATI, HTN
XPO inhibitor	Selinexor	Volume depletion	Hemodynamic AKI, hyponatremia
Immunotherapies			
CAR-T therapy	Axicabtagene ciloleucel, Idecabtagene vicleucel, brexucabtagene autoleucel, Tisagenlecleucel.	Systemic hyperinflammatory state. Ischemic injury	ATI, hemodynamic mediated AKI
CTLA-4 inhibitor	Ipilimumab, tremelimumab	Tubular injury, endothelial injury, podocyte injury	ATIN, ATI, MCD, lupus-like GN, necrotizing GN, TMA
PD-1 inhibitors	Pembrolizumab, cemiplimab, nivolumab	Tubular injury, endothelial injury, podocyte injury	ATIN, ATI, MCD, IgA nephropathy, FSGS, Necrotizing GN, Amyloidosis, Immune-complex mediated GN, C3 glomerulonephropathy
PDL-1 inhibitor	atezolizumab, avelumab, durvalumab	Tubular injury	ATIN, ATI

ALK, Anaplastic lymphoma; *ATI*, acute tubular injury; *ATIN*, acute tubulointerstitial nephritis; *ATP*, adenosine 5′-triphosphate; *BCL-2*, B cell lymphoma 2 gene; *BCR-ABL*, breakpoint cluster region-tyrosine protein kinase ABL-1 gene; *BRAF*, proto-oncogene B-Raf; *BTK*, Bruton tyrosine kinase; *CAR-T*, chimeric antigen receptor T cells; *CDK4/6*, cyclin-dependent kinase 4/6; *CTLA-4*, cytotoxic T-lymphocyte–associated protein 4; *DCT*, distal convoluted tubule; *EFGR*, epidermal growth factor receptor; *ERK*, extracellular signal-regulated kinase; *FSGS*, focal segmental glomerulosclerosis; *HTN*, hypertension; *MATE*, multidrug and toxin extrusion; *MCD*, minimal change disease; *MET*, mesenchymal-epithelial transition; *mTOR*, mammalian target of rapamycin; *NDI*, nephrogenic diabetes insipidus; *PARP*, poly-ADP-ribose polymerase; *PD-1*, programmed cell death protein 1; *PDL-1*, programmed death-ligand 1; *SIADH*, syndrome of inappropriate antidiuretic hormone secretion; *TLS*, tumor lysis syndrome; *TMA*, thrombotic microangiopathy; *VEGF*, vascular endothelial growth factor; *XPO*, export protein exportin 1.

Table_57_10

Table 57.11 Anti-VEGF Signaling Pathway Agents Kidney Adverse Events

Anti-VEGF Agents Subclasses	Drugs	Indications (Approved and Potential)	Proteinuria (Risk Factors: CKD, HTN, RCC)	Hypertension (Risk Factors: Age >60, BMI >25 kg/m^2, >1 Anti-VEGF Agent)
VEGF-Trap (fusion protein)	Alfibercept	Melanoma, prostate cancer, pancreatic cancer, non–small lung cancer	Incidence of 40%-60%	Incidence about 40%
VEGF-A monoclonal antibody	Bevacizumab	Glioblastoma, metastatic colorectal cancer, non–small cell lung cancer, renal cell carcinoma	Incidence of 20%-60%	Incidence about 20%
VEGFR2 monoclonal antibody	Ramucirumab	Metastatic colorectal cancer, breast cancer, non–small lung cancer	Incidence about 10%	Incidence about 42%
Tyrosine kinase inhibitors	Axitinib	Renal cell carcinoma, thyroid cancer, hepatocellular cancer, gastrointestinal stromal tumors, pancreatic neuroendocrine tumors, soft tissue sarcomas, refractory chronic myelogenous leukemia, and refractory metastatic colorectal cancer	Incidence about 20%	Incidence about 40%
	Cabozantinib		Incidence about 22%	Incidence about 30%-60%
	Lenvatinib		Incidence about 11%	Incidence about 70%
	Nintedanib		n/a	n/a
	Pazopanib		Incidence about 13%	Incidence about 35%
	Regorafinib		Incidence about 7%	Incidence about 28%
	Sorafenib		Incidence about 11%	Incidence about 19%
	Cabozantinib		Incidence about 8%	Incidence about 20%
	Vandetanib		Incidence about 10%	Incidence about 30%
	Apatinib		Incidence about 15%	Incidence about 9%

BMI, Body mass index; *CKD,* chronic kidney disease; *HTN,* hypertension; *RCC,* renal cell carcinoma; *VEGF,* vascular endothelial growth factor.

Table_57_11